

Shruti Kumari

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EDUCATION

Indian Institute of Technology, Madras

Bachelor of Science in Data Science

Chennai, India

2022 – 2026

CGPA: 8.19/10

EXPERIENCE

Research Intern | The Stemonef Enterprises Pvt. Ltd.

Jan 2026 – May 2026

- Transformed 15+ research reports on STEM, Data Science, and Computer Science topics into structured insights, supporting technical analysis and decision-making.
- Developed and delivered 5+ technical presentations on research findings, industry trends, and emerging technologies, improving knowledge sharing across project teams.
- Contributed to 2 web application projects through technical research, information gathering, documentation, and evidence-based recommendations.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, Bash, Git, Docker, DSA

ML/DL: Scikit-learn, XGBoost, PyTorch, TensorFlow, Feature Engineering, Model Evaluation, Statistical Modeling

LLMs & Generative AI: RAG, Fine-Tuning, Hugging Face, Prompt Engineering, LangChain, LangGraph, smolagents

MLOps & Deployment: FastAPI, Docker, MLflow, Airflow, DVC, GitHub Actions, CI/CD, Model Serving

Cloud Vector Databases: GCP, ChromaDB, FAISS

PROJECTS

Astraeus Multi-Agent RAG Platform

Astraeus Repository

- Architected a production-ready multi-agent Retrieval-Augmented Generation (RAG) platform using LangGraph, orchestrating specialized agents for memory management, document retrieval, research, tool execution, and response verification.
- Processed 50+ PDF documents and indexed 3,000+ semantic chunks in ChromaDB for semantic search, context-aware retrieval, knowledge discovery, and enterprise knowledge management.
- Built scalable backend services using FastAPI, enabling document upload, real-time query processing, streaming responses, API-driven inference, and conversational context management through session-based memory architecture.
- Integrated local Ollama LLMs, vector search, and retrieval workflows to deliver grounded answers from enterprise documents, reducing hallucinations through retrieval augmentation, response validation, and a dedicated verification layer.

ThreatGuard AI – Cyber Threat Intelligence Platform

ThreatGuard AI Repository

- Developed an end-to-end AI-powered cybersecurity platform combining Machine Learning, RAG, Agentic AI, and LLM-based threat analysis workflows for automated threat detection, investigation, and response assistance.
- Developed an XGBoost-based intrusion detection model on the CICIDS2017 dataset, achieving 97.05% classification accuracy across multiple cyber attack categories through feature engineering and data preprocessing.
- Built an intelligent RAG pipeline using FAISS, Sentence Transformers, and local Ollama LLMs to generate contextual threat investigation reports, IOC analysis, threat intelligence summaries, risk assessments, and mitigation recommendations.
- Engineered a full-stack solution using FastAPI and Next.js for real-time threat analysis, interactive security dashboards, API-driven inference, security visualization, and operational monitoring workflows.

MCP Incident Analyst – Agentic AI RCA Platform

MCP Incident Analyst Repository

- Built an Agentic AI incident investigation platform using LangGraph, Ollama, FastAPI, PostgreSQL, and ChromaDB, enabling automated Root Cause Analysis (RCA) through a 4-agent workflow.
- Developed AI agents for log analysis, GitHub commit correlation, historical incident retrieval, and RCA generation by analyzing 5+ recent code commits and operational evidence.
- Implemented RAG with ChromaDB and PostgreSQL persistence, supporting investigation history, report storage, and retrieval through 6+ REST API endpoints.
- Added OpenTelemetry tracing, structured logging, and execution metrics, generating end-to-end RCA reports in approximately 20–25 seconds using a local Ollama-hosted LLM.

CORE COMPETENCIES

Multi-Agent AI Systems • Retrieval-Augmented Generation (RAG) • LLM Applications • LangGraph Workflows • MLOps & Model Serving • End-to-End ML Pipelines • Vector Search & Embeddings • FastAPI Development • Cloud Infrastructure (GCP) • CI/CD Automation • Feature Engineering • Applied Machine Learning